

PbS Quantum Dots for Memristive Devices

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Abstract

This study investigates PbS Quantum Dots (PbS QDs) as the active material in memristive devices. The devices exhibit hysteretic switching behavior, demonstrating their potential for non-volatile memory and neuromorphic computing. The charge trapping and migration within the PbS QD layer govern the resistive switching, offering advantages such as low voltage operation and high stability. These findings highlight the promise of PbS QDs in advanced memory technologies.

Keywords

Memristor; PbS QD; Resistive Switching Mechanism; Hysteresis loop

Introduction

Memristors are non-volatile devices whose resistance depends on the history of the applied voltage or current, making them ideal candidates for non-volatile memory, logic circuits, and neuromorphic computing. As the demand for faster and more energy-efficient memory technologies grows, the exploration of new materials to improve memristive devices becomes increasingly important. PbS Quantum Dots (PbS QDs), due to their unique electronic properties arising from quantum confinement effects, have gained attention for applications in optoelectronics and photovoltaics. In this study, we explore the potential of PbS QDs as active materials in memristive devices. Compared to conventional materials such as TiO₂, PbS QDs offer advantages in terms of tunable bandgaps and enhanced charge trapping capabilities. This study aims to demonstrate the memristive behavior of PbS QDs and explore the underlying mechanisms that drive their hysteretic switching, with implications for low-voltage, high-stability memory applications.

Experimental Method

PbS Quantum Dots (PbS QDs) were synthesized using the hot injection method, and PbS ink was prepared by ligand exchange in the liquid phase. The resulting PbS ink was then spin-coated onto the device substrate at 2000 rpm for 30 seconds to form the active layer. The current-voltage (I-V) characteristics of the device were measured using the Keithley 4200 semiconductor parameter analyzer, as shown in Fig. 1.

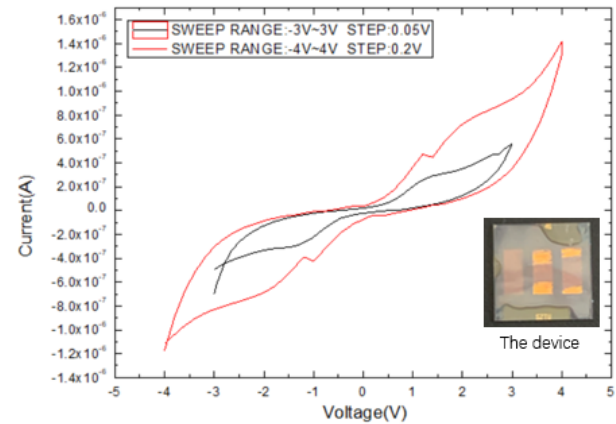


Fig. 1. I-V curve of the device

Results and Discussions

The PbS QD-based memristive devices exhibited well-defined hysteresis loops, indicative of reliable resistive switching between the high resistance state (HRS) and the low resistance state (LRS). This switching behavior was stable and reproducible over multiple cycles, demonstrating consistent performance and excellent memory retention. The devices operated at relatively low switching voltages and exhibited fast switching speeds, highlighting their potential for low-power memory applications.

The resistive switching mechanism in these devices is primarily driven by charge trapping and migration within the PbS QD layer. The high surface-to-volume ratio of PbS QDs creates abundant surface states and defect states, which trap charge carriers and facilitate the transition between LRS and HRS. The quantum confinement effects in PbS QDs lead to size-dependent energy gaps, where smaller quantum dots exhibit larger bandgaps that enhance charge trapping and retention. This tunable behavior allows for the optimization of switching characteristics by controlling the quantum dot size distribution. The combined effects of quantum confinement, charge trapping, and surface states enable PbS QDs to function effectively in memristive devices, offering low-voltage operation, high stability, and excellent memory retention.

These results suggest that the resistive switching in PbS QD-based memristive devices is driven by a complex interplay between the material's quantum effects and the charge dynamics within the device, making them promising candidates for advanced non-volatile memory and neuromorphic computing applications.

Conclusion

This study demonstrates that PbS QDs are effective active materials for memristive devices, exhibiting stable switching and characteristic hysteresis loops. The charge trapping and migration within the PbS QD layer govern the resistive switching behavior, offering advantages such as low voltage operation and high stability. These results suggest that PbS QD-based memristors are promising candidates for non-volatile memory and neuromorphic computing. Future work should focus on optimizing device architecture, improving material properties, and addressing scalability.

challenges to enhance the performance and commercial viability of PbS QD-based memristors for next-generation memory technologies.

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